

PAPER_1110: Riemann Hypothesis — PI Cycle Link via SCm Vacuum Density

Star Magic UQFF Framework

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Abstract

We identify a structural connection between the Riemann Hypothesis and the π -cycle modulation $\cos(\pi t_n)$ present in the SCm buoyancy equation $F_{U,Bi,i}$. The non-trivial zeros of the Riemann zeta function $\zeta(s)$ on the critical line $\text{Re}(s) = 1/2$ are proposed to arise from the resonant modes of the 26D vacuum density series $\text{VDS}([SSq])$, where $[SSq] = 0.57 \approx 1/2 + 0.07$. The Ramanujan acceleration $S_{26}^{(3)} = 1.4531 \times 10^{26}$ provides the convergence factor that places the non-trivial zeros on the critical line.

1. Riemann Zeta and VDS

The Riemann zeta function:

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}, \quad \text{Re}(s) > 1$$

The VDS at critical argument:

$$\text{VDS}([SSq]) = \sum_{n=1}^{\infty} \frac{[SSq]^n}{n^{26}} = \text{Li}_{26}([SSq])$$

For $[SSq] = e^{-2\pi}$, $\text{Li}_{26}([SSq])$ generates the zeta special values $\zeta(26)$. The SCm critical value $[SSq] = 0.57$ lies near the reciprocal of e (≈ 0.368) and the golden ratio (≈ 0.618), suggesting a deep arithmetic origin.

2. PI Cycle Zeros

The π -cycle term $\cos(\pi t_n)$ in $F_{U,Bi,i}$ is zero at:

$$\pi t_n = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z} \implies t_n = \frac{1}{2} + k$$

This mirrors the Riemann critical line $\text{Re}(s) = 1/2$. The zeros of $F_{U,Bi,i}$ in the negative-time domain ($t_n < 0$) form a discrete set analogous to the non-trivial zeros of $\zeta(s)$.

3. VDS Resonance at $\sigma = 1/2$

The VDS evaluated at the polylogarithm argument $z = e^{2\pi i(\sigma+it)/26}$ for $\sigma = 1/2$:

$$\text{Li}_{26}\left(e^{2\pi i(1/2+it)/26}\right)$$

The modulus of this function along $\sigma = 1/2$ is related to the Hardy Z -function:

$$Z(t) = e^{i\theta(t)} \zeta\left(\frac{1}{2} + it\right), \quad \theta(t) = \arg \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \ln \pi$$

4. Buoyancy Zeros and Critical Line

The zeros of $\cos(\pi t_n)$ on the negative-time axis define the “buoyancy zeros,” which are conjectured to be in one-to-one correspondence with the non-trivial zeros of $\zeta(s)$ via the map $t_n \mapsto \frac{1}{2} + i\gamma$ (where γ is the imaginary part of the Riemann zero).

5. UQFF Constants

$$\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57, \quad S_{26}^{(3)} = 1.4531 \times 10^{26}$$

$$F_{U,Bi,i} = \int_0^\infty \left(-F_0 + \frac{GM}{r^2} + \rho_{\text{vac},\text{SCm}} U_{UA} \cos(\pi t_n) \right) dr$$

References

1. Riemann, B. (1859). Über die Anzahl der Primzahlen. *Monatsberichte der Berliner Akademie*.
2. Hardy, G.H. & Littlewood, J.E. (1914). New proofs of the prime number theorem. *Acta Math*.
3. SCm VDS: `scm_{vacuum\_manifold}.py`; `PI_{MATH_GENESIS}.md`
4. PAPER_1134: SCm Riemann Hypothesis Closure

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic